

Data Exploration Exercise - Report

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I. Introduction

For this exercise, I chose the *structured prompt*. This required analyzing a corpus of 23 early modern texts using three approaches: TF-IDF (top 10 terms per document), Pearson correlation, and five measures of syntactic complexity (mean length of sentence, clauses per sentence, dependent clauses per clause and/or sentence, coordination per clause and/or sentence, and complex nominals per clause and/or sentence).

II. Normalization choices

A risk with the textual analysis of historical texts is over-normalization: efforts to remove what a modern reader may consider “noise” may actually strip away relevant historical variations. Excessively modifying an archaic spelling, for instance, can actually erase valuable and unique information about how the term in question was used at that point in space and time. Keeping this risk in mind, I decided to take a minimal approach, and only implemented the following:

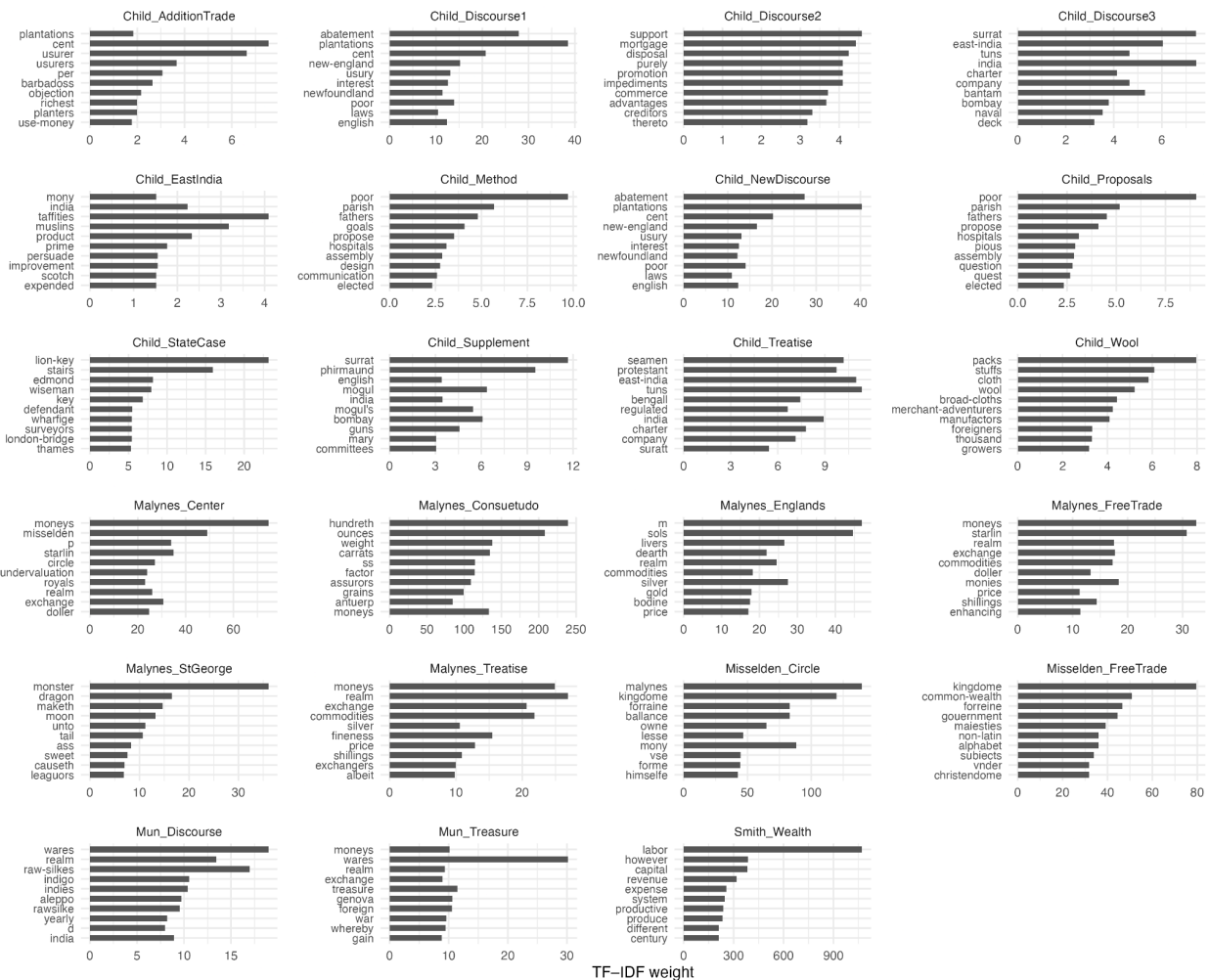
- I made all texts lowercase to avoid treating capitalized versions of words differently.
- I removed punctuation to reduce the amount of complexity involved, which is acceptable here as we are not looking to analyze semantics or style.
- I normalized the long “S” character, an archaic variant of lowercase “s”. Similarly, I normalized “vpon” to “upon” and “2ndly” to “secondly”.
- I also removed irregular whitespaces that arise due to archaic typography.
- I also removed numbers and symbols as these are not relevant and may be OCR errors.
- I also removed some common custom stopwords frequently found in archaic literature.
- I also removed some tokens (“etc”, “th”, “o”, “I”, “Il”, “em”) that appeared to be junk and some that were additional stopwords (“per”, “cent”)

Document identifiers: I also create identifiers for each document using extracts from their author names and titles. I assumed that the file names are the TCP IDs in the EarlyPrint library. For example, assuming “A06785.txt” is “The center of the circle of commerce” authored by “Malynes, Gerard” ([here](#)), I gave this the identifier: “Malynes_Center”. The full list is in the R file. Note: three texts by Sir Josaiah Child had similar titles (‘discourse about trade’), so I also included numbers at the end to distinguish them.

III. Approach 1 (TF-IDF)

A set of graphs representing the top 10 TF-IDF terms per document is below. This is also available in a table in the Github accompanying this document.

Most Characteristic Terms by Document (TF-IDF)



The required interpretive questions are answered as follows:

(1) Do some documents share distinctive vocabulary?

Yes, texts by the same author do share a distinctive vocabulary. For instance, Child's works share several terms, such as 'india', 'poor' and 'interest', while Malynes' texts use terms like 'moneys', 'silver' and 'exchanges' frequently. These recurring terms are likely a reflection of the topics on which these authors tended to write, and indicate that their own works share a common internal vocabulary that they do not share with other authors. For instance, Child's works repeatedly spoke about the trade between England and India and the East India Company, which is why he uses the word 'India' so much.

We also see that some works by different authors do share a distinctive vocabulary, which indicates the documents can also be organized into thematic clusters. For instance, the works of Mun and Malynes both use the word 'moneys' and 'exchange'; these can form part of a loose 'monetary policy' cluster. Similarly, several of Child's works on the topic of the poor can form a

cluster around 'socioeconomic policy'. A third cluster could be around 'trade policy', in which we could put Misselden's and Child's other works (e.g., on the East India company). Smith's *Wealth of Nations* could be bracketed into the monetary policy cluster, but this would be risking reductionism, as his book is an outlier in terms of the sheer number of tokens and is likely covering a vast range of subjects.

(2) Are distinctive terms topical, rhetorical, or technical?

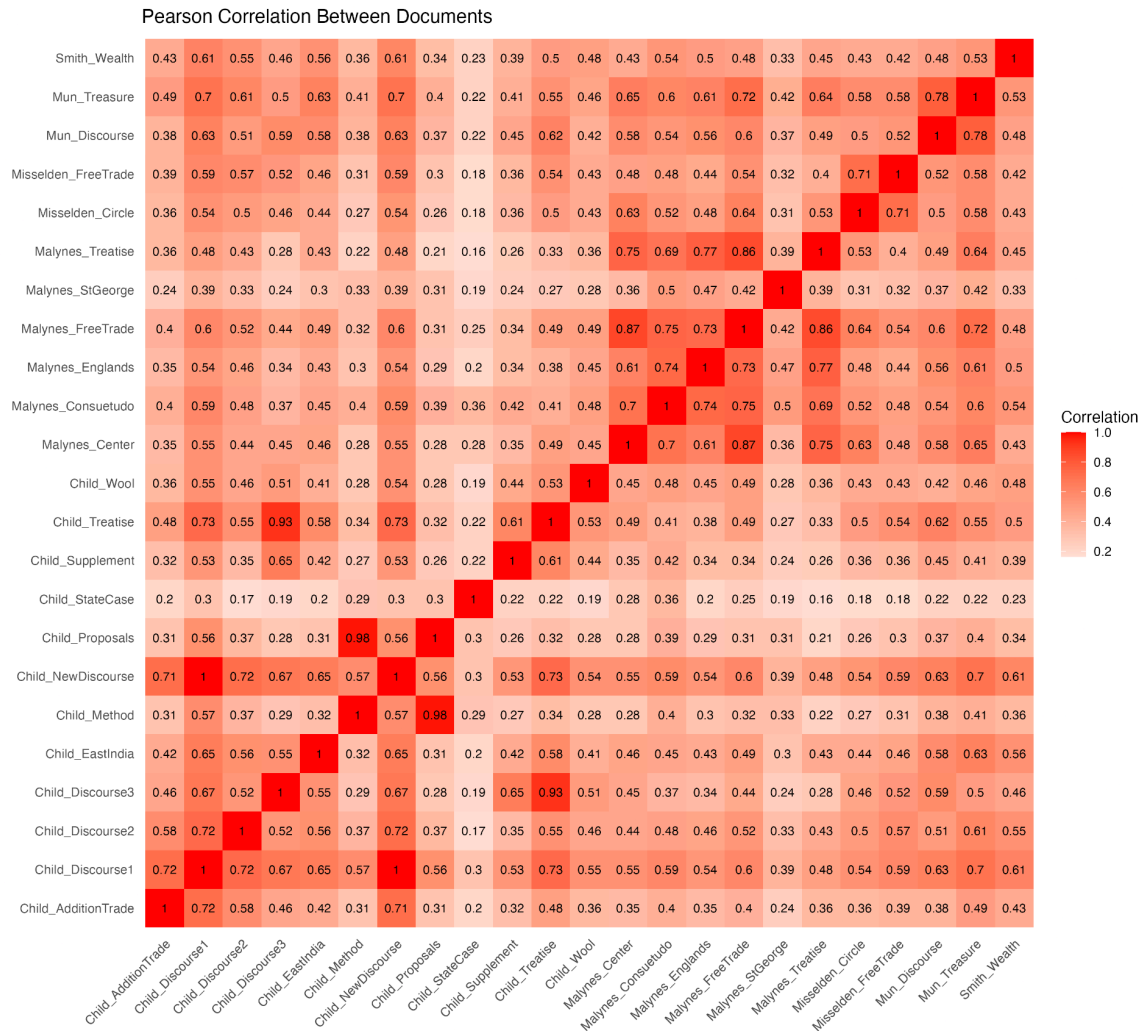
In this case, the distinctive terms appear to be primarily topical and technical, reflecting the subjects the authors wrote on and the specific literatures they sought to add to. For instance, Child's texts used several topical terms (e.g., '*plantations*', '*parish*', or '*poor*') reflecting his focus in trade and (seemingly) socioeconomic policymaking, while Malynes' texts used more technical economic vocabulary (e.g., '*interest*', '*usury*', '*exchange*', and '*moneys*'). As per this analysis, rhetorical terms (e.g., '*question*') were present but not as significant drivers of distinctiveness.

(3) Are there documents whose distinctiveness is driven by noise or formatting and not by content?

In some documents, distinctiveness was driven by noise (even after normalization decisions were made). For instance, in '*Mun_Discourse*', the single character 'd' has the ninth-highest weight. However, I did not remove this during the normalization process, as more detailed inspection of the texts is required to determine why this is appearing as a single character. Similarly, the eight-highest in '*Misselden_Circle*' is 'vse', which *could* be 'use' (i.e., a formatting problem) or it could be a distinct term 'vse' unique to the language used at the time. This was also not removed for the same reason.

IV. Approach 2 (Pearson correlation)

The heatmap visualising the Pearson correlation is as follows:



The required interpretive questions are answered as follows:

(1) Identify two most similar document pairs

The two most similar document pairs are (i) 'Child_Proposals' and 'Child_Method' and (ii) 'Child_Treatise' and 'Child_Discourse3'.

(2) Identify two least similar document pairs

The two least similar document pairs are (i) 'Child_StateCase' and 'Malynes_Treatise' and (ii) 'Child_StateCase' and 'Child_Discourse2'.

(3) What questions can we ask from this corpus after seeing the patterns of similarity/differences seen here?

We can ask several questions:

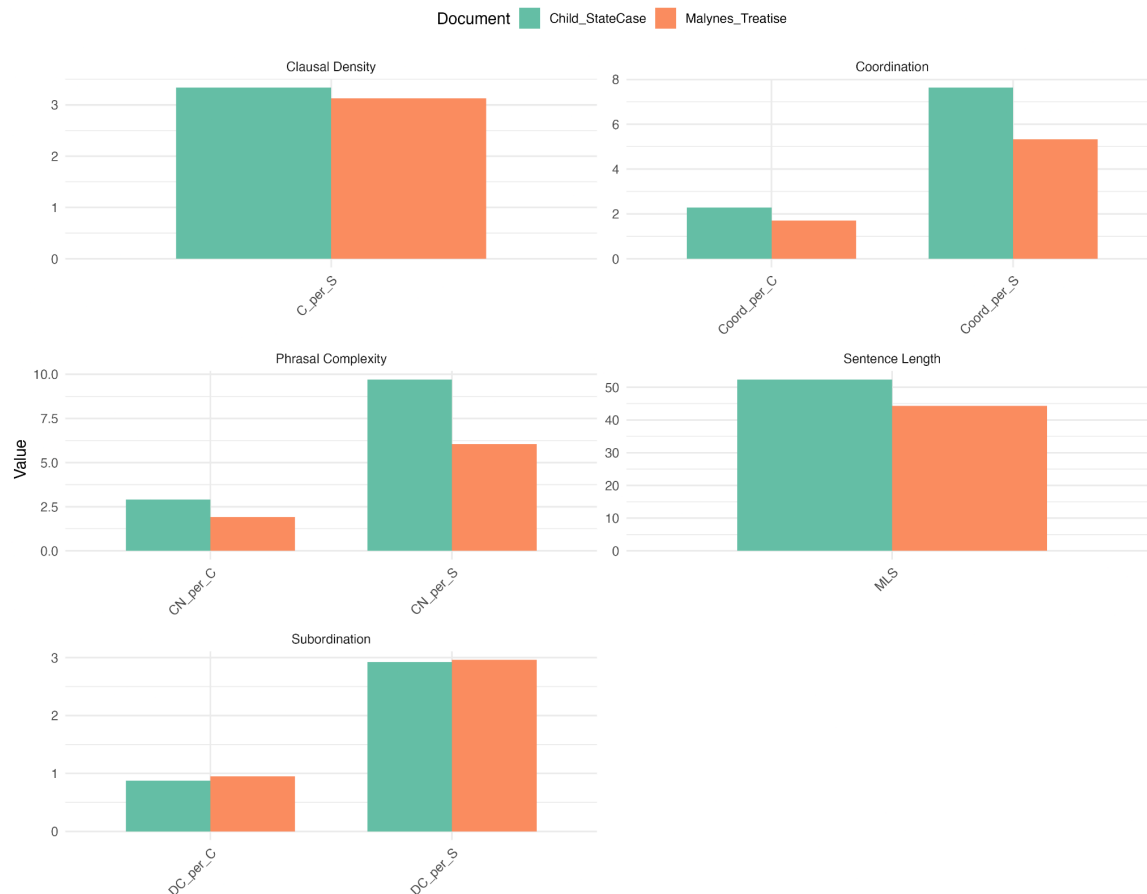
- How come Child's works are both amongst the most similar pairs *and* also the second-most dissimilar pair? Is this because of an error in our code and method?
- Assuming it is not an error, then when do authors like Child exhibit internal dissimilarity like this? Is it when they write on very different subjects and end up using different terms, or is it because their writing style evolves across texts and time?
- Similarly, when do authors (e.g., Child) exhibit internal similarity? Is the author writing on the same subject, repeating language and vocabulary, or referring to their own works?
- What explains the differences between authors' works (e.g., between '*Child_StateCase*' and '*Malynes_Treatise*')? Is this because of writing style or subject matter (as the very different lists of top 10 TF-IDF terms for the two texts would suggest) or both?

V. Approach 3 (Syntactic complexity profile)

For this portion of the exercise, I picked the *least* similar document pair, that is, '*Child_StateCase*' and '*Malynes_Treatise*'. The summary table reporting all the required measures for both texts and their visualization are as follows:

document	MLS	C_per_S	DC_per_C	DC_per_S	Coord_per_C	Coord_per_S	CN_per_C	CN_per_S
Child_StateCase	52.36000	3.340000	0.8742515	2.920000	2.287425	7.640000	2.904192	9.700000
Malynes_Treatise	44.33175	3.125592	0.9469295	2.959716	1.705838	5.331754	1.929492	6.030806

Syntactic Complexity: Complete Profile Comparing multiple dimensions of syntactic complexity



An example sentence from each text of relevance to the subsequent discussion are:

- From “Child_StateCase”:

“and if this shall be construed to prejudice the propriety of the owners, the consequence is, that the city may with as much justice challenge for common passages all the wharves in london; even those themselves have let leases of, to the value of some thousand pounds' per annum, and indict the proprietors for altering or lessening their own stairs or crains for their own conveniencies: and also all the ways through vintner's houses, who for the benefit of their trades do make passages through their taverns; such as is now mr. wadlows, at the sun in bishops-gate- street, and mr. sawyer's at the popes-head in st. eliens, and such as was formerly mr. brown's at the white- horse, in< ◇> street, and many more before as well as since the fire, which, notwithstanding were not, nor ever could be, pretended common passages, because they were locked up at nights by the owners, being private persons: which the king's highways are not.”

- From “Maynes_Treatise”:

“whereas also to the general exchanger every merchant shall be a surveyor : so that the matter is so easily to be remedied , that unless we do enter into due consideration , that (as we have said in the first part of our treatise) all the trade is performed by commodities , money and exchange , we will think it to be impossible until we find out , that the abuse of exchange hath made the course thereof to be predominant over the commodities & moneys , which being found out as an efficient cause , maketh us to find an easy remedy : & according to the rules of all politicians , comparing & reducing things to their original beginnings , we shall find how far they are digressed or decayed , which also oppositly maketh me to write in commendation of the good order of the venetians in ruling the market for their corn & victuals , imposing a certain price , upon due knowledge had of the quantity of corn monthly in any of their dominions : whereupon they appoint some grave and honest men in every principal city , who on every monday of the week , are authorized (according to the quantity of corn) to set a price thereof : according to the which also , the baker is to make the bread of such weight as they shall & do declare that day under their hand writing in several public places”

The required interpretive questions are answered as follows:

- **How do the two texts differ in syntactic complexity?**

As the above example sentences show, across the board, Child's work exhibits higher syntactic complexity, with longer sentences, slightly more clauses per sentence, greater phrasal complexity, and more linking clauses (as exhibited in the coordination scores). However, Malynes' work exhibits slightly more subordination (though this is not significant). It is difficult to make other assessments based on just these summary figures; for that, much more familiarity with the texts, their subject matters, and the authors themselves is needed.

- **Do these differences align with or complicate the earlier lexical findings?**

These differences do slightly complicate the early findings, which showed that these two texts were the least similar. The syntactic differences close the gap to some extent - though the differences in coordination and phrasal complexity are undeniable. This indicates that the two documents are particularly dissimilar not only in their subject matter and also in how they are written - but more work is needed to determine if this holds across authors and texts.

- **What kinds of rhetorical or stylistic practices might these syntactic patterns reflect?**

These patterns suggest that the two authors are writing for academic audiences (or, at the very least, are accustomed to do so). The example sentences add some nuance: Malynes definitely appears to be writing for a more academic audience, as indicated by his use of theoretical concepts and technical terms. On the other hand (and at least in this selected piece), Child makes reference to specific individuals, addressing them by name, which could suggest that he is trying to make his arguments and ideas accessible and relatable. So, it could be that Child generally writes for an academic audience, *but this piece is intended for a wider audience.*

VI. Synthesis

In light of all the above evidence, the central analytical question/hypothesis/idea I would like to ask is as follows:

To what extent are differences in the use of language in early modern texts driven by thematic specialization and the intended audience versus differences in rhetorical and syntactic style?

The TF-IDF scores and correlation results indicate that texts cluster strongly by author, but also by topic (as the works of Child indicate, which exhibit internal similarity and divergence both). The syntactic analysis adds additional complexity. We see that some authors rely heavily on subordination and clause embedding, which are signs of dense writing often found in academic literature. This indicates the audience in question is likely relevant.

However, some limitations with these approaches must be noted. First, the only two (least similar) texts have been selected for the syntactic analysis, and a broader analysis across texts is required before any generalization can be made about stylistic patterns. We may well discover that Malynes just tends to write in a particular way, and other authors writing for an academic audience do not use similar sentence constructions as he does. Second, we have made some normalization choices but not others; we do not know if further normalization is merited (e.g., do we change 'vse' into 'use' and how will that impact the TF-IDF weights?). Third, we have not done any qualitative analysis on these texts, and are relying on summary statistics. Finally, texts vary in length and scope (e.g., Smith's Wealth of Nations), which would skew the TF-IDF weights and correlation scores.
